Statement of Work (SOW)

1. Background

NASA Glenn Research Center (GRC) requires a **large periphery model** and **Gallium Nitride (GaN) diode devices** to support ongoing research and simulation efforts. These components are essential for accurate modeling, circuit prototyping, and advancing related technology development.

2. Scope

The contractor shall provide:

- A **large periphery model** that is **compatible with Cadence software** or an equivalent industry-standard platform capable of supporting the required simulation and design work.
- **Gallium Nitride (GaN) diodes** that meet the following specifications:
  - **Part Number:** C170581-J-03 CHIP-20 GaN Schottky diodes
  - **Quantity:** [200]
  - **Performance requirements:**
    - global cut-off frequency (f_{-3dB}) > 30 GHz
    - Junction Capacitance < 1 pf at 10 GHz
    - Operational power handling of > 2W of incident, continuous wave, power through the diode at 10 GHz
    - Bare die with GSG probe pads and top microstrip gold plates contacts

3. Technical Requirements

- The periphery model must be **fully compatible with NASA GRC's existing simulation environment**, ensuring seamless integration with ongoing **BEAM simulation efforts**.
- The GaN diodes must match the specified part number to maintain consistency in **circuit performance, testing, and prototyping**.
- Alternative models or components must be **validated for compatibility** before acceptance.

4. Deliverables

- Delivery of the **large periphery model** in a format compatible with NASA's design software.
- **GaN diodes** delivered within the required timeframe to support project milestones.
- Documentation detailing the specifications, usage guidelines, and compatibility verification.

5. Period of Performance

The required components shall be delivered no later than **7/2025** to prevent delays in research and development.

The models should be delivered no later than **10/2025** to prevent delays in research and development.

6. Government Furnished Information (GFI)

NASA will provide the necessary simulation parameters and design specifications to ensure proper integration of the components.